NTG Case

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English Translation

1. Executive case summary

Working title

NTG Pulse

Subtitle: Verified Freight Visibility for UK - Nordics ROAD Corridors

Brief core of the case

NTG Pulse is an event-based visibility layer for ROAD shipments. The solution uses strategic freight gates to create verified milestone events that can improve ETA confidence, reduce manual tracking inquiries, and strengthen NTG's digital differentiation in complex transport corridors.

The solution should not be understood as "license plate tracking". It should be understood as a verified freight milestone layer on top of existing tracking data.

Sharp formulation

NTG Pulse does not try to track every road. It verifies the checkpoints that matter.

Main point

Large visibility platforms often estimate shipment progress based on GPS, telematics, carrier data, and geofencing. NTG Pulse adds an infrastructure-based truth layer where selected gates confirm that a shipment has actually passed critical checkpoints.

2. Problem definition

The overall problem

ROAD visibility is often fragmented, delayed, and dependent on data sources that NTG does not fully control.

Typical data sources:

- GPS/telematics
- carrier systems
- ELD/tachograph
- driver apps
- TMS/ERP events
- EDI statuses
- manual emails/calls

Why this creates problems

B2B customers expect:

- reliable shipment status
- mobile-friendly tracking
- predictable ETA
- proactive delay communication
- fewer manual touchpoints

However, ROAD transport often has:

- carrier handoffs
- different systems
- driver-managed devices
- missing or delayed GPS data
- uncertainty around ports, tunnels, ferries, and borders
- manual updates

Consequences for NTG

- more "where is my shipment?" inquiries
- lower ETA confidence
- more internal follow-up
- poorer customer experience
- weaker digital differentiation in tenders
- difficulties in complex UK/EU/Nordics corridors

Problem statement

NTG lacks a more reliable visibility layer for complex ROAD corridors, where existing tracking is often fragmented, delayed, or dependent on carrier and driver behavior.

3. Solution concept

Basic idea

When a registered shipment passes an approved gate, an event is generated.

Example:

```
{
"event_id": "evt_10091",
"shipment_id": "SHP-2026-00421",
"gate_owner": "Scandlines",
"gate_id": "RODBY_EXIT",
"gate_type": "FERRY_TERMINAL",
"event_type": "GATE_PASSED",
"plate_hash": "8f41ab92...",
"equipment_id": "TRAILER-NTG-4412",
"timestamp": "2026-05-07T11:42:00Z",
"direction": "INBOUND_DK",
"confidence_score": 0.94,
"source": "SIMULATED_ANPR",
"status": "MATCHED"
}
```

This event is linked to an active shipment and displayed as a verified milestone.

Example shipment timeline

Manchester -> Copenhagen

- Loaded in Manchester
- Checked in at Dover/Folkestone
- Arrived at Calais/Coquelles
- Passed through German corridor
- Checked in at Puttgarden
- Exited at Rødby
- Arrived Copenhagen terminal

What the solution does in practice

- confirms physical passage through important chokepoints
- updates the shipment timeline
- improves ETA confidence
- detects mismatches and route deviations
- sends uncertain events to manual review
- reduces the need for manual status follow-up
- gives customers a more trustworthy tracking picture

4. What the product is not

NTG Pulse is not:

- live video surveillance
- access to public cameras
- mass monitoring of license plates
- a national ANPR platform
- a replacement for GPS
- an attempt to build project44/FourKites/Shippeo from scratch
- a solution that requires new cameras everywhere

NTG Pulse is:

- event-only
- shipment-specific
- whitelist-based
- asset-light
- partner-driven
- corridor-focused
- a supplement to existing GPS/telematics/TMS data

5. Product architecture

1. Gate Event Simulator / Gate Owner Event Source

In the PoC, gate events are simulated. In a real pilot, events would come from gate owners such as DFDS, Scandlines, Stena Line, LeShuttle, Portbase, or similar actors.

2. Event Ingestion API

Receives events as JSON, EDI, CSV, XML, webhook, or SFTP feed.

For the PoC: JSON via REST API.

For real operation: EDI, SFTP, and partner APIs are likely.

3. Identity & Match Layer

Matches the event against an active shipment.

Matches on:

- shipment ID
- hashed plate ID
- trailer ID
- equipment ID
- booking reference
- carrier reference

4. Validation Engine

Checks whether the event is trustworthy.

Validates:

- does the shipment exist?
- is the shipment active?
- does the plate/equipment match?
- is the gate expected on the route?
- is the timestamp realistic?
- is the direction correct?
- is the confidence score high enough?
- is the carrier/trailer still linked to the shipment?

5. Milestone Engine

Translates technical events into customer-friendly milestones.

Examples:

- Checked in at ferry terminal
- Departed UK
- Arrived in EU
- Entered Denmark corridor
- Passed final checkpoint
- Delivered

6. ETA & Confidence Engine

Updates ETA and confidence score based on the latest verified event, historical route, and expected next checkpoint.

7. Exception Engine

Finds problems.

Examples:

- missing expected gate event
- route deviation
- low confidence match
- impossible travel time
- unexpected gate
- no update for X hours

8. Customer Portal

Customer-friendly view.

Should show:

- shipment timeline
- last verified checkpoint
- ETA
- confidence status
- delay warning
- next expected checkpoint

9. Internal Control Tower

Internal dashboard.

Should show:

- shipments at risk
- delayed shipments
- missing milestone events
- mismatch queue
- manual review queue
- customer impact priority

10. Governance & Compliance Layer

Handles:

- audit trail
- retention
- access control
- data minimization
- purpose limitation
- DPIA documentation
- controller/processor agreements

6. Data types and integration formats

Typical gate systems can deliver data in several ways.

REST API / JSON

The most modern and easiest to use in the PoC. Suitable for events such as:

- vehicle_checked_in
- unit_loaded
- gate_passed
- departed
- arrived
- released

EDI

Very common in transport and logistics. Used for:

- bookings
- status messages
- customs
- consignment notes
- terminal events

CSV / Excel export

Practical and realistic for an initial PoC or batch import.

XML

Still seen in enterprise and logistics systems, especially older integrations.

Webhook events

A modern setup where the partner sends events directly to NTG at check-in, boarding, arrival, etc.

SFTP file exchange

Very common in enterprise logistics. The partner places EDI/CSV/XML files in a folder, and NTG retrieves them periodically.

ANPR event payloads

Specifically from license plate recognition systems.

Typical fields:

- number plate or hashed plate
- timestamp
- gate ID
- camera ID
- confidence score
- direction
- event ID
- image reference, if allowed

Recommendation

The PoC should use JSON via REST API.

Real operation should be able to support EDI, SFTP, and partner APIs.

7. Technical PoC

Purpose of the PoC

The PoC should not prove that NTG can build ANPR cameras. It should prove that gate events can be converted into useful shipment visibility.

Simple PoC architecture

Gate Event Simulator -> Event Ingestion API -> Validation Engine -> Shipment Visibility Service -> Dashboard / Map / Timeline

Fixed demo gates

Recommended 3-5 gates:

- DOVER_CHECKIN or FOLKESTONE_CHECKIN
- CALAIS_EXIT or COQUELLES_EXIT
- PUTTGARDEN_CHECKIN
- RODBY_EXIT
- CPH_TERMINAL_ARRIVAL

Confidence score

Confidence should be shown as a decision status, not just a number.

Score	Status	Meaning
0.90-1.00	Confirmed	Secure match
0.70-0.89	Needs review	Likely match, requires validation
Below 0.70	Rejected	Too uncertain for automatic tracking

Demo display

Example:

```
Scandlines Rødby Exit
Matched trailer: TRAILER-NTG-4412
Confidence: 94%
Status: Confirmed
ETA updated: 12:55
```

Mismatch / false positive

Mismatch is handled through the validation pipeline.

The event is only accepted if:

- the shipment is active
- equipment matches
- the gate matches the route
- the timestamp is realistic
- direction makes sense
- confidence is above the threshold

Three outcomes

Auto-accept

High confidence + route/timing OK.

Manual review

Medium confidence or minor deviation.

Reject

Low confidence, unknown shipment, wrong gate, or physically impossible event.

Example mismatch

```
{
"event_id": "evt_10097",
"shipment_id": "SHP-2026-00421",
"gate_id": "STOREBAELT_EAST",
"confidence_score": 0.76,
"status": "NEEDS_REVIEW",
"reason": "Gate not expected on planned route"
}
```

Carriers with changing tractors/trailers

The PoC must not only track the tractor. That would be too naive in international transport.

The system must separate:

- shipment
- carrier
- tractor/truck
- trailer
- container/load unit
- plate IDs
- gate events

Important principle decision

Track shipment through equipment association, not only truck plate.

Example

```
Shipment SHP-00421
-> Trailer ID TR-4412
-> Tractor Plate AB12345 until Dover
-> Tractor Plate CD67890 after Calais
-> Carrier: Partner Carrier A
```

Equipment handover event

```
{
"shipment_id": "SHP-2026-00421",
"event_type": "EQUIPMENT_REASSIGNED",
"old_tractor_plate_hash": "old123...",
"new_tractor_plate_hash": "new456...",
"trailer_id": "TRAILER-NTG-4412",
"timestamp": "2026-05-07T08:30:00Z",
"reason": "Carrier handover at Calais"
}
```

Route deviation

Route deviation does not automatically mean an error. It means that the shipment is no longer following the expected route assumption.

Three levels

Type	Example	Action
Minor deviation	Late arrival at expected gate	Update ETA
Alternative route	Hoek van Holland instead of Calais	Recalculate route + flag
Critical deviation	Gate in wrong country/direction	Manual review / exception

Example

```
{
"shipment_id": "SHP-2026-00421",
"event_type": "ROUTE_DEVIATION",
"expected_gate": "CALAIS_EXIT",
"actual_gate": "HOEK_VAN_HOLLAND_EXIT",
"severity": "MEDIUM",
"action": "ETA_RECALCULATED",
"customer_visible": true
}
```

8. Minimum data model for the PoC

shipments

- shipment_id
- customer
- origin
- destination
- status
- planned_route_id
- current_eta

equipment_assignments

- shipment_id
- carrier_id
- trailer_id
- tractor_plate_hash
- valid_from
- valid_to

gates

- gate_id
- gate_owner
- location
- type

gate_events

- event_id
- shipment_id
- gate_id
- timestamp
- confidence_score
- status
- reason

route_plans

- route_id
- expected_gate_sequence

exceptions

- exception_id
- shipment_id
- type
- severity
- status

9. UK - Nordics corridor focus

Why UK/Nordics is strong

UK - Europe - Nordics is a complex ROAD corridor with many places where uncertainty arises.

Typical pains:

- Brexit/customs uncertainty
- port congestion
- ferry delays
- tunnel delays
- border friction
- carrier handoffs
- ETA volatility
- fragmented carrier data
- high customer anxiety

Strategic pitch angle

Where complexity is highest, visibility is worth the most.

Relevant corridor gates

Tier 1:

- Dover
- Eurotunnel / LeShuttle Freight
- Calais / Coquelles
- DFDS / ferry operator events

Tier 2:

- Harwich / Hoek van Holland
- Rotterdam / Portbase
- Immingham
- Felixstowe
- Puttgarden / Rødby

Tier 3:

- Padborg / Flensburg
- Storebælt
- Øresund
- Copenhagen terminal / customer site

10. Potential gate-owner partners

Primary PoC recommendation

Start with commercial freight actors, not public cameras.

Prioritized PoC package:

1. DFDS
2. Scandlines Freight
3. Stena Line Freight or LeShuttle Freight
4. Portbase/Rotterdam, if the route goes via Holland
5. Sund & Bælt as technically relevant, but possibly heavier as a first partner

1. DFDS

Very strong candidate.

Why:

- UK/EU routes
- Danish presence
- freight/logistics business
- digital freight tracking
- Enterprise EDI

Possible events:

- booking created
- booking status
- unit/trailer status
- check-in
- ferry departure
- ferry arrival
- port-to-port tracking

PoC relevance: very high.

2. Scandlines Freight

Very strong candidate for the Denmark/Germany corridor.

Why:

- Puttgarden-Rødby is a strong checkpoint before Denmark/Copenhagen
- Cargo Web
- Freight App
- EDI
- online booking for freight customers

Possible events:

- booking created
- booking changed
- truck/unit booked on sailing
- check-in
- boarding
- departure/arrival via booking status

PoC relevance: very high.

3. Stena Line Freight

Strong candidate, especially at Harwich-Hoek van Holland.

Why:

- Freight Portal
- tracking units
- incoming unaccompanied units
- units on quay
- SmartGate
- license plate linked to booking at terminal entrance

Possible events:

- booking created/updated
- unit checked in
- trailer drop-off
- trailer pick-up
- ready-for-pickup
- license plate linked to booking
- gate entry confirmation

PoC relevance: very high for the Holland route.

4. LeShuttle Freight / Eurotunnel

Strong candidate, but enterprise access may be heavier.

Why:

- controlled tunnel passage
- online booking
- vehicle tracking
- self check-in
- ANPR / number plate recognition
- automatic arrival notification

Possible events:

- vehicle pre-registered
- vehicle checked in
- number plate matched to account
- vehicle entered terminal
- vehicle crossed tunnel
- vehicle arrived other side

PoC relevance: high technical value, medium realistic access.

5. Portbase / Rotterdam

Strong candidate for the Holland route and terminal data.

Why:

- Port Community System
- terminal data via web/API
- integration with terminal planning systems

Possible events:

- container/unit status
- terminal status
- expected discharge time
- cargo release status
- import flow updates
- pre-notification

PoC relevance: high if the route goes via Rotterdam/Hoek van Holland.

6. Destin8 / MCP UK ports

Relevant for UK port events and customs/port-community flows.

Possible events:

- cargo arrival
- export/import status
- customs clearance status
- port release
- RoRo/unitised cargo movement
- manifest-linked events

PoC relevance: medium-high, but probably not the first integration.

7. P&O Ferries Freight

Relevant, but API/EDI evidence appears weaker than DFDS/Stena/Scandlines.

Possible events:

- booking created/amended
- unit tracking
- check-in status
- shipment updates
- notifications

PoC relevance: medium.

8. Irish Ferries Freight

Relevant, but secondary for England-Copenhagen.

Possible events:

- booking created
- booking managed
- shipment tracking
- route updates

PoC relevance: low-medium.

11. Gate access reality: Dover, Calais, Eurotunnel

Eurotunnel / LeShuttle Freight

Most likely direct data source.

Strengths:

- account-based online services
- vehicle tracking
- booking/API indications
- ANPR/self check-in
- arrival notifications

Assessment: very strong, but requires formal partner/account access.

Port of Calais

Realistic, but likely indirect.

Best access routes:

- ferry operators
- E-modal/terminal systems
- Port Community System

Assessment: strong as part of corridor proof, but not necessarily via a "Calais API".

Port of Dover

Strong physical gate point, but not obvious as a directly open API.

Best access routes:

- DFDS
- P&O Ferries
- Irish Ferries
- customs/GVMS-related systems
- special partnership

Assessment: medium. Use Dover as a physical gate in the demo, but be careful about claiming direct Port of Dover API access.

Sharp formulation

The PoC should not be based on an assumption of direct access to public port data. The realistic model is commercial event data through existing freight accounts and operator platforms.

12. NTG own sites and asset-light reality

NTG appears to have its own offices/sites on the UK/Nordics route, but NTG is primarily an asset-light freight forwarder, not a heavy terminal owner.

Relevant NTG points

- NTG Road UK Ltd., Felixstowe
- NTG Road UK Ltd., Lichfield / Midlands
- UK logistics hotspots
- customs / road freight / groupage / full load activity

Case consequence

NTG sites can be used as internal or partner-adjacent data points, but the strongest gates in the case remain ferry ports, tunnel passages, terminal operators, and infrastructure partners.

Important formulation

NTG does not need to own every gate. It needs to orchestrate verified events from the gates its shipments already pass.

13. Competitive analysis

The market

Real-time transportation visibility is dominated by large platforms such as:

- project44
- FourKites
- Shippeo
- Transporeon
- Descartes MacroPoint
- GoComet
- Locus
- FarEye
- Tive
- Samsara

These platforms primarily build on:

- GPS
- telematics
- ELD
- TMS/ERP integrations
- carrier APIs
- driver apps
- IoT devices
- geofencing

project44

Offers:

- real-time truck tracking
- carrier integrations
- GPS/telematics
- predictive ETA
- exception alerts
- geofencing
- multimodal visibility

Strengths:

- large integration network
- enterprise maturity
- global coverage
- predictive ETA

Weaknesses:

- dependent on carrier compliance
- GPS data can be fragmented
- not an independent infrastructure-based truth layer

FourKites

Offers:

- real-time shipment visibility
- dynamic ETA
- exception management
- predictive alerts
- geofencing
- operational orchestration
- digital twin/logistics intelligence

Strengths:

- strong analytics
- disruption management
- customer visibility

Weaknesses:

- still dependent on GPS/telematics
- tracking gaps at carrier handoffs
- black spots at borders, ports, and terminal changes

Shippeo

Offers:

- real-time FTL/LTL tracking
- predictive ETA
- exception management
- geofencing
- carrier onboarding
- TMS/ERP integration
- carbon visibility
- driver app support

Strengths:

- strong ETA engine
- large carrier network
- high tracking coverage
- good European ROAD relevance

Weaknesses:

- dependent on carriers and devices
- driver adoption can be critical
- not a checkpoint-verification layer as the core product

Transporeon

Offers:

- transport execution
- carrier management
- yard management
- dock scheduling
- visibility dashboards
- ETA prediction

Strengths:

- strong in Europe
- closely connected to enterprise transport flows

Weaknesses:

- more of an operations platform than a truth layer
- not a dedicated checkpoint-verification system

Descartes MacroPoint

Offers:

- GPS tracking
- freight visibility
- mobile tracking
- ETA updates
- geofencing
- shipment status updates

Weaknesses:

- heavily dependent on mobile/GPS tracking
- tracking gaps
- not infrastructure-based

Tive / Samsara / Locus / FarEye / GoComet

These are relevant, but not the main competitors in your narrow position.

- Tive: strong for IoT/high-value cargo
- Samsara: strong for fleet telematics
- Locus/FarEye: more last-mile/delivery orchestration
- GoComet: more freight procurement/shipment visibility dashboard

14. NTG Pulse vs. the competitors

The strategic distinction

Competitors typically ask:

Where do we estimate the shipment is?

NTG Pulse asks:

Which critical checkpoints has the shipment verified passed?

GPS/geofencing

Geofencing uses a virtual area around a location. When the GPS signal enters the area, an event is created.

Problems:

- GPS can be inaccurate
- data can be delayed
- carrier data can be missing
- driver apps can be offline
- GPS does not necessarily mean that the vehicle has actually checked in, loaded, crossed, or been released

NTG Pulse

Builds on verified checkpoint events from physical infrastructure:

- ports
- bridges
- ferries
- tunnels
- terminals
- gates

Sharp difference

Geofencing:

We think the truck is here.

NTG Pulse:

We know the shipment passed this checkpoint.

Positioning

NTG Pulse should not compete directly on global scale, carrier onboarding, or multimodal visibility.

It should be positioned as:

- corridor-specific visibility enhancement
- infrastructure-based truth layer
- verified milestone layer
- premium visibility feature
- NTG-owned differentiator

15. Build, buy, or partner?

Build everything internally

Advantages:

- full ownership
- unique differentiation
- data ownership

Disadvantages:

- expensive
- slow
- complex
- requires a large integration network
- high compliance burden

Assessment: NTG should not build the entire visibility stack itself.

Buy an existing platform

Advantages:

- fast implementation
- mature technology
- enterprise support

Disadvantages:

- low differentiation
- vendor dependency
- same solution as competitors

Assessment: useful as a base, but not enough as a unique NTG position.

Hybrid / partner-led architecture

Recommended model.

NTG buys/partner-integrates:

- GPS visibility
- telematics
- carrier data
- event infrastructure where possible

NTG builds:

- checkpoint intelligence layer
- verified milestone logic
- confidence engine
- corridor rules
- customer-facing NTG experience
- exception handling for high-value corridors

Conclusion

NTG Pulse should be a specialized layer on top of the existing visibility stack, not a replacement for it.

16. GDPR and legal model

Basic premise

License plates can be personal data when they can be linked to a driver, carrier, vehicle, booking, or shipment.

Therefore, the solution must be:

- data-minimized
- purpose-limited
- event-only
- whitelist-based
- transparent
- contract-regulated

Recommended legal PoC model

Scenario A: The gate owner scans and sends only match/no-match or shipment events to NTG.

The gate owner scans as usual, but NTG only receives an event if a whitelisted NTG shipment matches.

Roles

Gate owner:

- controller for the camera/ANPR system itself

NTG:

- controller for its own shipment/tracking data

The gate owner can also be a processor for NTG for the specific matching service, if it only matches NTG's whitelist according to NTG's instructions.

Why this model is strong

NTG does not receive raw plate data for all other trucks. NTG only receives events relevant to active shipments.

What NTG should receive

```
{
"shipment_id": "SHP-12345",
"hashed_plate_id": "abc123...",
"gate_id": "SCANDLINES_RODBY_IN",
"event_type": "GATE_PASSED",
"timestamp": "2026-05-07T10:32:00Z",
"confidence": 0.96
}
```

What NTG should not receive

- all plates seen at the gate
- raw camera images
- non-matching vehicles
- general traffic data
- searchable plate database

DPIA

A lightweight DPIA should be conducted for a pilot.

The DPIA must cover:

- purpose
- data minimization
- retention
- access control
- lawful basis
- risks for drivers
- how raw data is avoided or deleted quickly

Information to carrier/driver

Carrier is informed through:

- transport agreement
- partner agreement
- data appendix

Driver is informed through:

- the carrier's internal information
- booking/app flow
- driver instruction
- privacy notice
- signage at gate

Lawful basis

Legitimate interest is the most realistic primary basis for shipment tracking, combined with contracts/data agreements between NTG, carrier, and gate owner.

Consent should be avoided as the primary basis, because it is weak in employment/driver relationships and may make the system unstable.

Retention

There is no single fixed deadline. Retention must follow the purpose.

Recommended PoC model:

- non-matching plates are deleted immediately or within 24 hours by the gate owner
- matched NTG events are stored briefly for error handling/audit
- NTG does not receive images by default
- longer storage only occurs as a pseudonymized shipment event

Hashed plate/equipment IDs

NTG can likely store hashed plate/equipment IDs, but they are still personal data if NTG can link them to a shipment, vehicle, carrier, or driver.

Recommendation:

- store hashed_plate_id
- store equipment_id/trailer_id
- store shipment_id
- store gate_id
- store timestamp
- avoid raw plate in the primary tracking database
- store the mapping separately and access-restricted
- delete/rotate mapping after shipment or a short audit period

Short legal formulation

NTG Pulse should use whitelist-based gate matching where gate owners process raw ANPR data locally and only send verified shipment events to NTG. NTG does not need camera access or full plate databases.

17. Economic business case

What do we not know?

We do not know NTG's actual:

- cost per manual tracking inquiry
- number of WISMO/tracking inquiries per month
- inquiry rate per shipment
- support time per inquiry
- premium customers' willingness to pay
- tender/retention value

Therefore, the economics should be presented as scenarios, not as false precision.

WISMO cost estimate

Recommended standard estimate:

- DKK 75 per manual tracking inquiry

This corresponds approximately to 8-12 minutes of employee time including overhead.

Types of inquiries

Type of inquiry	Time	Estimated cost
Simple lookup	3-5 min	DKK 20-50
Normal email/call	8-12 min	DKK 60-120
Escalated inquiry	20-30 min	DKK 150-300

Inquiry-rate scenarios

Use 5-10% of relevant shipments as a conservative/medium estimate.

Shipments/month	5%	10%	20%
1,000	50	100	200
2,500	125	250	500
5,000	250	500	1,000
10,000	500	1,000	2,000

Payback scenario A: support savings only

Assumptions:

- 500 tracking inquiries per month
- DKK 75 per inquiry
- 40% reduction

Calculation:

- 500 x 75 x 40% = DKK 15,000/month
- annually: DKK 180,000

Pilot cost	Payback from support only
DKK 1 million	approx. 5.6 years
DKK 2 million	approx. 11.1 years
DKK 3 million	approx. 16.7 years

Conclusion A

Support savings alone are not enough to justify a DKK 1-3 million pilot.

Payback scenario B: support + premium visibility

Assumptions:

- support savings: DKK 180,000/year
- 5 premium customers
- DKK 10,000/month per customer

Premium revenue:

- 5 x 10,000 x 12 = DKK 600,000/year

Total annual value:

- DKK 780,000/year

Pilot cost	Payback
DKK 1 million	approx. 1.3 years
DKK 2 million	approx. 2.6 years
DKK 3 million	approx. 3.8 years

Payback scenario C: support + premium + tender/retention

Assumptions:

- support savings: DKK 180,000/year
- premium visibility: DKK 600,000/year
- one customer is won/retained because of visibility
- conservative annual contribution margin value: DKK 500,000

Total annual value:

- DKK 1,280,000/year

Pilot cost	Payback
DKK 1 million	approx. 0.8 years
DKK 2 million	approx. 1.6 years
DKK 3 million	approx. 2.3 years

Main economic conclusion

A pilot should not be sold as "we save customer service emails". That is too weak.

It should be sold as:

- premium visibility product
- tender differentiator
- retention driver
- operational automation
- support savings as a secondary benefit

18. Premium visibility customers

The most obvious customers are those for whom delay, uncertainty, or missing documentation costs money.

Most relevant customer types

Customer type	Why they would pay
Pharma / healthcare	temperature, compliance, high value
Food / chilled / frozen	temperature sensitivity and delivery windows
High-value electronics	theft risk and insurance
Automotive / spare parts	JIT flow; production stoppages are expensive
Retail / wholesalers	campaigns, inventory management, customer promises
Project cargo / special transport	complexity and coordination
Dangerous goods	compliance and risk management
4PL/control tower customers	they are specifically buying overview and reporting

Recommendation

Do not start with all customers. Start with:

- pharma/healthcare
- temperature-controlled goods
- high-value shipments
- automotive/JIT
- 4PL/control tower customers

19. Pilot scope

Recommended pilot

90 days

3-5 gates

1 specific UK - DK/Nordics route

Proposed PoC route

England -> Dover/Folkestone -> Calais/Coquelles -> Germany -> Puttgarden -> Rødby -> Copenhagen

Data points

- UK exit
- EU entry
- Denmark approach
- Danish arrival corridor
- destination terminal

Primary KPIs

- inquiry reduction
- ETA accuracy
- ETA-confidence improvement
- customer portal usage
- number of auto-confirmed events
- number of manual reviews

- number of false positives rejected
- route deviation detection
- customer satisfaction / perceived visibility
- internal time saved

20. Answered questions v2

Which specific gate owners could be realistic first partners?

Answered. The strongest candidates are DFDS, Scandlines, Stena Line, LeShuttle Freight, and Portbase/Rotterdam. DFDS + Scandlines are most obvious as the first PoC package.

Does NTG have its own terminals or fixed partner sites on the UK/Nordics route?

Partially answered. NTG has UK sites/logistics hotspots, including Felixstowe and Lichfield, but should be understood as an asset-light freight forwarder. Own sites can be internal data points, but not the main gate infrastructure.

Which ports/ferries/tunnels have existing commercial event/API opportunities?

Answered at case level. DFDS, Scandlines, Stena Line, LeShuttle, and Portbase have relevant digital freight/booking/tracking/EDI/API-like opportunities.

Can Dover/Calais/Eurotunnel offer data through commercial partnerships?

Partially answered. Eurotunnel/LeShuttle appears strongest as a direct candidate. Calais is realistic via ferry operators/PCS. Dover is most likely realistic via ferry operators/customs/GVMS/special partnership.

Which data formats can typical gate systems deliver?

Answered. REST/JSON, EDI, CSV, XML, webhook, SFTP, and ANPR event payloads.

Who is data controller and processor?

Answered for the recommended PoC scenario. The gate owner is controller for camera/ANPR, and NTG is controller for shipment tracking. The gate owner may be processor for the matching service, depending on the agreement.

Should a DPIA be prepared?

Answered. Yes, recommend a lightweight DPIA for the pilot.

How are carrier/driver informed?

Answered. Carrier via contract/data appendix. Driver via carrier information, booking/app flow, privacy notice, and gate signage.

Can legitimate interest be used?

Answered as a recommendation. Yes, probably best as the primary basis, combined with contracts/data agreements. Consent should be avoided as the primary basis.

How long may raw plate data be stored?

Partially answered. No fixed deadline. The PoC should use the shortest possible retention: non-matches deleted immediately/within 24 hours, matched events stored briefly for audit/errors.

May NTG store hashed plate/equipment IDs?

Answered. Yes, but they are still pseudonymized personal data if NTG can link them back to shipment/vehicle/driver.

What does a manual WISMO inquiry cost NTG?

Partially answered with an estimate. NTG data is missing, but DKK 75 per inquiry can be used as a conservative case estimate.

How many tracking inquiries does NTG receive per month?

Not answered precisely. Requires internal NTG data. Can be modeled with a 5-10% inquiry rate of relevant shipments.

How much value does better visibility have in tenders?

Partially answered. Cannot be quantified externally, but should be argued as tender score, retention, SLA documentation, and premium service.

Which customers would pay for premium visibility?

Answered. Pharma, chilled/frozen, high-value, automotive/JIT, dangerous goods, project cargo, and 4PL/control tower customers.

What is the payback period for a DKK 1-3 million pilot?

Answered as scenarios. Support alone is weak. With premium visibility and retention/tender value, payback can realistically be around 1-3 years.

How should the PoC simulate gate events?

Answered. Gate Event Simulator sends synthetic JSON events to the ingestion API.

How should the demo show confidence score?

Answered. As Confirmed / Needs Review / Rejected with percentage and explanation.

How should mismatch/false positive be handled?

Answered. Validation engine with route, timing, active shipment, equipment match, and threshold.

How are carriers with changing tractors/trailers handled?

Answered. Track shipment through equipment assignments, especially trailer/load unit, not just truck plate.

How is route deviation handled?

Answered. Planned route is compared with observed gate events. Deviations are classified as minor, alternative route, or critical deviation.

21. Still missing / should be investigated

Internal NTG data

- actual tracking inquiries per month
- support time per tracking inquiry
- number of shipments on UK/DK/Nordics corridor
- existing TMS/CRM/ticket system
- current carrier integrations
- customer segments with the greatest visibility need

Partner reality

- whether DFDS/Scandlines/Stena/LeShuttle will actually share events commercially
- price per integration
- whether API/EDI access requires account, volume, or enterprise agreement
- data ownership in partner agreements

Legal

- concrete DPIA
- concrete lawful basis assessment
- controller/processor agreement per partner
- retention policy per data source
- driver information in practice

Technical

- actual data model in NTG's existing systems
- how events are linked to shipments
- whether trailer ID or booking reference is the most stable identifier
- whether customer portal or internal dashboard should be the primary demo

Economics

- premium pricing validation
- customers' willingness to pay
- cost per gate integration
- support reduction measurement
- tender win-rate/retention effect

22. Q&A-ready answers

How do you get access to data?

We do not start with public cameras. We start with commercial freight partners that already have booking, check-in, terminal, or gate events. DFDS, Scandlines, Stena Line, LeShuttle, and Portbase are examples of realistic data sources. Data should be delivered as event output via API, EDI, webhook, SFTP, or CSV - not as live video.

What about GDPR?

The model is based on whitelist-based matching. The gate owner handles raw ANPR data locally and only sends verified shipment events to NTG on match. NTG does not receive all license plates or camera feeds. For a pilot, a lightweight DPIA, clear retention, access control, and carrier/driver information should be prepared.

Why not just GPS?

GPS is useful, but depends on carriers, devices, integrations, and data quality. NTG Pulse does not replace GPS. It adds verified milestones from physical infrastructure, so NTG can say "the shipment actually passed this checkpoint", not just "the GPS signal was nearby".

How is this different from geofencing?

Geofencing creates events based on GPS position in a virtual area. NTG Pulse uses actual gate events from controlled checkpoints. Geofencing estimates. Gate events verify.

What does it cost?

A serious pilot is still estimated at around DKK 1-3 million, depending on integrations, legal/compliance, product development, and partner agreements. The PoC can be built more cheaply by simulating gate events and focusing on event logic, validation, and dashboard.

Is payback realistic?

Not if the business case is only based on fewer tracking emails. Support savings alone give weak payback. With premium visibility, tender differentiation, and retention, a DKK 1-3 million pilot can realistically give payback around 1-3 years in a positive scenario.

Who would pay for premium visibility?

Customers with high risk from delay or low tolerance for uncertainty: pharma, chilled/frozen, high-value goods, automotive/JIT, dangerous goods, project cargo, and 4PL/control tower customers.

Should NTG build everything itself?

No. The most realistic strategy is hybrid. NTG can buy or integrate existing GPS/visibility data, but build checkpoint intelligence, confidence scoring, corridor rules, and customer experience on top.

How do you handle false positives?

All events go through a validation engine that checks active shipment, equipment match, expected route, timing, direction, and confidence score. Events are either confirmed, sent to manual review, or rejected.

How do you handle changing tractors/trailers?

The system tracks shipment through equipment assignments, especially trailer/load unit, not just truck plate. If the tractor changes at Calais but the trailer continues, shipment tracking can continue.

How is route deviation handled?

Observed gate events are compared with the planned route. Minor deviations update ETA, alternative routes are flagged and recalculated, and critical deviations are sent to manual review.

23. Final case positioning

NTG Pulse should be positioned as a verified corridor intelligence layer.

This means:

- not another GPS platform
- not pure ALPR
- not surveillance
- not a national camera solution
- not a replacement for project44/FourKites/Shippeo

But:

- a specialized layer on top of existing visibility
- focused on high-value ROAD corridors
- driven by verified freight events
- built for ETA confidence, exception handling, and premium customer visibility

Final pitch formulation

NTG Pulse combines existing transport visibility with verified checkpoint events from strategic freight gates. The result is stronger confidence, fewer blind spots and a premium customer experience on complex UK - Nordics ROAD corridors.